EXPERIMENT 3

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D20A-02

Prompt Engineering Using Large Language Models

AIM:

To design and evaluate effective prompts using Zero-shot, Few-shot, Chain-of-Thought (CoT), and Role Prompting techniques for improving the quality, accuracy, and relevance of responses generated by Large Language Models (LLMs).

THEORY:

Prompt Engineering is the process of designing and structuring instructions given to Large Language Models (LLMs) to obtain accurate, relevant, and meaningful outputs. Large Language Models such as GPT, Gemini, Claude, and Llama are trained on massive amounts of text data and can perform various Natural Language Processing tasks, including text generation, summarization, translation, question answering, code generation, and reasoning. These models generate responses based on the input prompt provided by the user. Therefore, the quality of the prompt plays an important role in determining the quality of the generated response.

A prompt generally consists of instructions, context, examples, and the expected output format. A clear and specific prompt helps the model understand the task and reduces ambiguity. Prompt Engineering allows users to control the behavior of an LLM without modifying or retraining its underlying parameters.

There are several prompting techniques used to guide LLMs:

1. Zero-shot Prompting:

Zero-shot prompting involves asking the model to perform a task without providing any examples. The model relies entirely on its previously learned knowledge to understand the instruction and generate an answer. For example, asking an LLM to summarize an article in 100 words without showing it a sample summary is zero-shot prompting. It is useful for simple and commonly understood tasks.

2. Few-shot Prompting:

Few-shot prompting provides the model with a small number of input-output examples before asking it to perform a similar task. These examples help the model understand the expected format, style, and pattern of the response. For example, providing two examples

of sentiment classification before asking the model to classify a new sentence helps improve consistency.

3. Chain-of-Thought Prompting:

Chain-of-Thought prompting encourages the model to approach a complex problem through intermediate reasoning steps before presenting its final answer. It is useful for mathematical problems, logical reasoning, and multi-step tasks. This technique helps structure the problem-solving process and can improve performance on tasks that require multiple operations.

4. Role Prompting:

Role prompting assigns a particular role or identity to the LLM before giving it a task. For example, instructions such as "You are an experienced software engineer" or "You are a career counselor" guide the model to generate responses appropriate to that role. It helps produce more focused and domain-specific answers.

The performance of these techniques can be evaluated by comparing generated responses based on accuracy, completeness, clarity, relevance, creativity, reasoning quality, response length, and hallucinations. Hallucinations refer to information generated by the model that may be incorrect or unsupported.

By applying different prompting techniques to the same tasks, the influence of prompt structure on the output can be studied. This experiment helps understand how effective prompt design improves the interaction between users and generative AI systems.

CODE AND OUTPUT:

Experiment 3 - Prompt Engineering using Large Language Models

```
!pip -q install transformers sentencepiece torch pandas
```

```
import pandas as pd
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
import torch, re, textwrap, time

MODEL_NAME = "google/flan-t5-small"
device = "cuda" if torch.cuda.is_available() else "cpu"
tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME)
model = AutoModelForSeq2SeqLM.from_pretrained(MODEL_NAME).to(device)
model.eval()
print("Model:", MODEL_NAME, "| device:", device)
```

```
config.json: 100% 1.40k/1.40k [00:00<00:00, 31.6kB/s]
tokenizer_config.json: 100% 2.54k/2.54k [00:00<00:00, 79.1kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
spiece.model: reconstructing file: 100% 792kB / 792kB, 78.0kB/s
spiece.model: downloading bytes: 649kB, 63.9kB/s
tokenizer.json: 100% 2.42M/2.42M [00:00<00:00, 15.9MB/s]
special_tokens_map.json: 100% 2.20k/2.20k [00:00<00:00, 28.9kB/s]
model.safetensors: reconstructing file: 100% 308MB / 308MB, 27.7MB/s
model.safetensors: downloading bytes: 291MB, 24.9MB/s
Loading weights: 100% 190/190 [00:00<00:00, 578.81it/s]
[transformers] The tied weights mapping and config for this model specifies to tie shared.weight to lm_head.weight, but both are present in the checkpoints with different values,
generation_config.json: 100% 147/147 [00:00<00:00, 1.71kB/s]
Model: google/flan-t5-small | device: cpu
```

```
def generate(prompt, max_new_tokens=100):
    inputs = tokenizer(prompt, return_tensors="pt", truncation=True, max_length=512).to(device)
    with torch.no_grad():
        out = model.generate(**inputs, max_new_tokens=max_new_tokens, num_beams=2)
    return tokenizer.decode(out[0], skip_special_tokens=True)

tasks = [
    {
        "task": "Explain a concept",
        "input": "Explain why a cricket batter should keep their head balanced while playing a straight drive.",
        "few": "Input: Explain why a batter watches the ball.\nOutput: Watching the ball helps the batter judge its line and length and time the shot.\n\nInput: Explain why footwork matters.\nOutput: Footwork helps the batter reach the ball while maintaining balance.\n\nIn
    },
    {
        "task": "Create a drill",
        "input": "Suggest a safe 10-minute beginner drill to practise front-foot defence.",
        "few": "Input: Give a 5-minute warm-up.\nOutput: 1) Easy jogging, 2) shoulder mobility, 3) gentle dynamic stretches. Stop if pain occurs.\n\nInput: Suggest a catching drill.\nOutput: Use soft throws, start close, focus on watching the ball, and increase distance gr
    },
    {
        "task": "Summarize",
        "input": "Summarize in two bullets: A batter should assess the line and length, move their feet early, keep their head steady, and practise consistently. A coach should give specific feedback and avoid changing too many things at once.",
        "few": "Input: Summarize: Warm up first, practise slowly, then increase pace.\nOutput:\n- Warm up before practice.\n- Increase intensity gradually.\n\nInput: Summarize: Watch the ball and keep balance.\nOutput:\n- Track the ball.\n- Maintain balance.\n\nInput: " +
    },
    {
        "task": "Give feedback",
        "input": "A fictional batter repeatedly loses balance after attempting a cover drive. Give three constructive coaching suggestions without diagnosing an injury.",
        "few": "Input: A batter swings too early.\nOutput: 1) Track the ball longer. 2) Practise slowly with controlled feeds. 3) Ask a coach to observe timing.\n\nInput: A batter's backlift is inconsistent.\nOutput: 1) Use shadow practice. 2) Repeat a comfortable setup. 3
    },
    {
        "task": "Answer with uncertainty",
        "input": "Can a webcam alone prove that a cricket batter will score a century in their next match? Answer briefly and explain the limitation.",
        "few": "Input: Can a video guarantee a player will never miss a shot?\nOutput: No. Video can show technique in the recorded conditions, but it cannot guarantee future performance.\n\nInput: Can one practice session prove long-term improvement?\nOutput: No. Repeated
    }
]

techniques = ["Zero-shot", "Few-shot", "Step-by-step", "Role"]
records = []
```

```python
techniques = ["Zero-shot", "Few-shot", "Step-by-step", "Role"]
records = []
for task in tasks:
    prompts = {
        "Zero-shot": task["input"],
        "Few-shot": task["few"],
        "Step-by-step": "Answer the task. Give a concise numbered explanation of the main considerations, then a final answer. Do not invent facts.\nTask: " + task["input"],
        "Role": "You are a patient cricket batting coach who gives practical, safe, beginner-friendly advice. Be specific, concise, and state uncertainty when needed.\nTask: " + task["input"],
    }
    for technique in techniques:
        answer = generate(prompts[technique], max_new_tokens=100)
        records.append({"Task":task["task"], "Technique":technique, "Prompt":prompts[technique], "Output":answer,
                        "Word count":len(answer.split())})
        print(f"\n[{task['task']} | {technique}]\n{answer}")

results_df = pd.DataFrame(records)
display(results_df[["Task","Technique","Output","Word count"]])
```

```
[Explain a concept | Zero-shot]
Cricket batters should keep their head balanced while playing a straight drive.

[Explain a concept | Few-shot]
Output: Cricket batters should keep their head balanced while playing a straight drive.

[Explain a concept | Step-by-step]
A straight drive is a way to keep your head balanced while playing a straight drive.

[Explain a concept | Role]
If you are a patient cricket batting coach, it is important to keep your head balanced while playing a straight drive.

[Create a drill | Zero-shot]
Using a hammer drill, drill a hammer into a hammer.

[Create a drill | Few-shot]
Output: Use a drill that is easy to use and easy to use.

[Create a drill | Step-by-step]
The first drill is a drill that is designed for beginners. The first drill is a drill that is designed for beginners. The second drill is a drill that is designed for beginners. The third drill is a drill that is designed for beginners. The fourth drill is a drill that i

[Create a drill | Role]
If you are a cricket player, you should be able to do it with a batting coach.

[Summarize | Zero-shot]
Assess the line and length of the batter.

[Summarize | Few-shot]
Output: Identify the line and length of the batter.
```

```
[Summarize | Role]
Be patient. Be patient.

[Give feedback | Zero-shot]
Using a cover drive is a good way to improve your balance.

[Give feedback | Few-shot]
Output: 1) Ask a coach to help him with his recovery. 2) Ask a coach to help him with his recovery. 3) Ask a coach to help him with his recovery.

[Give feedback | Step-by-step]
A fictional batter loses balance after attempting a cover drive. A fictional batter loses balance after attempting a cover drive. A fictional batter repeatedly loses balance after attempting a cover drive. A fictional batter repeatedly loses balance after attempting a cove

[Give feedback | Role]
Make a list of the three suggestions.

[Answer with uncertainty | Zero-shot]
Cricket is a game that is played on a webcam. Cricket is a game that is played on a webcam. Cricket is a game that is played on a webcam. Cricket is a game that is played on a webcam. Cricket is a game that is played on a webcam. Cricket is a game that is played on a webcr

[Answer with uncertainty | Few-shot]
Output: No. A webcam is a device that can be used to watch cricket matches. Cricket is a sport that can be used to watch cricket matches. Cricket is a sport that can be used to watch cricket matches. Cricket is a sport that can be used to watch cricket matches. Cricket is

[Answer with uncertainty | Step-by-step]
A webcam is a device used to measure the speed of a cricket match. The webcam is a device used to measure the speed of a cricket match. It is a device used to measure the speed of a cricket match. The webcam is a device used to measure the speed of a cricket match. The web

[Answer with uncertainty | Role]
If you are a cricket batting coach, you should be able to use a webcam to prove that a cricket batter will score a century in their next match. If you are a cricket batting coach, you should be able to use a webcam to prove that a cricket batter will score a century in the
```

	Task	Technique	Output	Word count
0	Explain a concept	Zero-shot	Cricket batters should keep their head balance...	12
1	Explain a concept	Few-shot	Output: Cricket batters should keep their head...	13

	Task	Technique	Output	Word count
0	Explain a concept	Zero-shot	Cricket batters should keep their head balance...	12
1	Explain a concept	Few-shot	Output: Cricket batters should keep their head...	13
2	Explain a concept	Step-by-step	A straight drive is a way to keep your head ba...	16
3	Explain a concept	Role	If you are a patient cricket batting coach, it...	21
4	Create a drill	Zero-shot	Using a hammer drill, drill a hammer into a ha...	10
5	Create a drill	Few-shot	Output: Use a drill that is easy to use and ea...	13
6	Create a drill	Step-by-step	The first drill is a drill that is designed fo...	66
7	Create a drill	Role	If you are a cricket player, you should be abl...	17
8	Summarize	Zero-shot	Assess the line and length of the batter.	8
9	Summarize	Few-shot	Output: Identify the line and length of the ba...	9
10	Summarize	Step-by-step	Observe the batter's line and length. Observe ...	46
11	Summarize	Role	Be patient. Be patient.	4
12	Give feedback	Zero-shot	Using a cover drive is a good way to improve y...	12
13	Give feedback	Few-shot	Output: 1) Ask a coach to help him with his re...	31
14	Give feedback	Step-by-step	A fictional batter loses balance after attempt...	73
15	Give feedback	Role	Make a list of the three suggestions.	7
16	Answer with uncertainty	Zero-shot	Cricket is a game that is played on a webcam. ...	72
17	Answer with uncertainty	Few-shot	Output: No. A webcam is a device that can be u...	83
18	Answer with uncertainty	Step-by-step	A webcam is a device used to measure the speed...	79
19	Answer with uncertainty	Role	If you are a cricket batting coach, you should...	58

```
ratings = results_df[["Task","Technique","Output"]].copy()
for col in ["Correctness (1-5)", "Relevance (1-5)", "Completeness (1-5)", "Clarity (1-5)", "Hallucination observed?"]:
    ratings[col] = ""
display(ratings)

ratings.to_csv("experiment3_prompt_evaluation.csv", index=False)
```

	Task	Technique	Output	Correctness (1-5)	Relevance (1-5)	Completeness (1-5)	Clarity (1-5)	Hallucination observed?
0	Explain a concept	Zero-shot	Cricket batters should keep their head balance...					
1	Explain a concept	Few-shot	Output: Cricket batters should keep their head...					
2	Explain a concept	Step-by-step	A straight drive is a way to keep your head ba...					
3	Explain a concept	Role	If you are a patient cricket batting coach, it...					
4	Create a drill	Zero-shot	Using a hammer drill, drill a hammer into a ha...					
5	Create a drill	Few-shot	Output: Use a drill that is easy to use and ea...					
6	Create a drill	Step-by-step	The first drill is a drill that is designed fo...					
7	Create a drill	Role	If you are a cricket player, you should be abl...					
8	Summarize	Zero-shot	Assess the line and length of the batter.					
9	Summarize	Few-shot	Output: Identify the line and length of the ba...					
10	Summarize	Step-by-step	Observe the batter's line and length. Observe ...					
11	Summarize	Role	Be patient. Be patient.					
12	Give feedback	Zero-shot	Using a cover drive is a good way to improve y...					
13	Give feedback	Few-shot	Output: 1) Ask a coach to help him with his re...					
14	Give feedback	Step-by-step	A fictional batter loses balance after attempt...					
15	Give feedback	Role	Make a list of the three suggestions.					
16	Answer with uncertainty	Zero-shot	Cricket is a game that is played on a webcam. ...					
17	Answer with uncertainty	Few-shot	Output: No. A webcam is a device that can be u...					
18	Answer with uncertainty	Step-by-step	A webcam is a device used to measure the speed...					
19	Answer with uncertainty	Role	If you are a cricket batting coach, you should...					

https://colab.research.google.com/drive/1Iz5c-jMIXrVOFAC8RozprT32OU-yhZn8?usp=sharing

CONCLUSION:

The experiment demonstrated the use of different prompt engineering techniques, including Zero-shot, Few-shot, Chain-of-Thought, and Role Prompting. The generated responses were compared to understand how prompt structure influences accuracy, relevance, clarity, and completeness. It was observed that different prompting techniques are suitable for different tasks, and an appropriately designed prompt can significantly improve the quality of LLM-generated responses.